Where Does Cold Dark Energy Switch On? A Precisely Posed Question at $M \gtrsim 10^{12.4}\,M_\odot$

Working note, 2026-10-27, v1.0. AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ is fitted; the theory is not closed.

Abstract

The programme studies a galaxy law, $g_N = g^2/(g + a_0)$ with $a_0 = c^2\sqrt{\Lambda/32\pi}$ ($\kappa = \frac{1}{2}$ fitted), together with a gravitating $w \approx 0$ field component, called here *cold dark energy* (not matter, not a particle). The law fits isolated galaxies; groups and clusters need cold dark energy halos. The record's best organising rule is environment-keyed: cold dark energy halos in satellites and in systems above a mass threshold; the law on baryons alone below it. This note does not claim a mechanism. It states, from frozen and reproduced calculations, *what the mechanism must do*:

1. **The threshold is not the core-floor crossover.** If the threshold is X43's crossover $\eta\,\Sigma_{\rm coll}(M) = a_0/2\pi G$ (with $\eta \propto c_f^{5/4}$), one number c_f sets both where condensation starts and how concentrated group and cluster cores are. On a grid $c_f = 3.5$–6.0 no value passes every graded line (CF1). KiDS isolated-lens lensing passes only at $c_f = 3.5$, i.e. a threshold $\geq 10^{12.39}\,M_\odot$ (canonical footing); groups pass only for $c_f \geq 4.75$ on both footings, which lowers the threshold below $10^{11.72}$.
2. **The cluster halo budget is set by measured gas.** With the record's Burkert halos, cluster outskirts need $c_f \geq 5.25$ and centres $c_f \leq 5.0$. The tension dissolves when the halo is built from measured inputs: counting the X-COP-measured gas (density slope $s = 2.777$ beyond R_{500}, 11 clusters) out to the self-consistent turnaround radius and assigning the native fair share of cold dark energy, all 15 eRASS1×KiDS cluster keys pass at the external $c_f = 4$ on both footings (GS1; its frozen verdict is still a fail, because of the group line; KiDS lensing and the dwarfs are not yet regraded at this budget). Correlated structure beyond turnaround contributes at most 1.3 % (GC1); a self-similar infall shape fixes the cluster centres (GC3).
3. **What remains of the cores is one line.** At the measured slope the groups need $c_f \gtrsim 4.25$–4.75 while the clusters take $c_f \approx 4$; the record's native zooms give c_f falling with mass, the direction required.
4. **Below the threshold**, the surviving exclusion mechanism (M6, initial displacement) is not yet tested by the Lyman-α forest: four native forest models of increasing fidelity are each not diagnostic. If the forest-side budget is real, galaxies needing M6 must keep at least 19–56 % (canonical) or 28–83 % (alternative) of their initial patch baryons, against the record's own Lagrangian retention of 19.1 % / 16.3 %.

The questions posed: what physics switches cold dark energy on at $M \gtrsim 10^{12.4}\,M_\odot$, independently of the concentration of the halos it then forms; and what makes groups more concentrated than clusters?

1 Setting and rules

- $\kappa = \frac{1}{2}$ is fitted. Footings: canonical $a_0 = 9.3603 \times 10^{-11}$, alternative $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, never pooled.
- Cold dark energy: a gravitating $w \approx 0$ field component paired with dark energy; never a particle. ΛCDM appears only as the rival.
- Every lane: frozen criteria committed alone before any number, a Step-0 reproduction stop rule, a mutation twin, one thread, under 1 GB. Inputs are raw data or framework-native; the native background is h 0.6856427, Ω_m 0.2976016, ω_b 0.0223081 (BAO+BBN, no CMB).
- Nothing here is evidence for the framework over ΛCDM.

2 The organising rule and its threshold

X58 established that the required retention of cold dark energy is non-monotone in mass: Local Group dwarfs (mostly satellites) retain, isolated $L_\star$ galaxies, the SPARC large spirals and the Milky Way disc exclude it, and groups and clusters retain it. X62 encoded this as an environment-keyed switch: a system carries a route-E halo if it is a satellite or lies above X43's crossover; otherwise the law acts on its baryons alone. The crossover is the mass at which the infall-concentrated collapse surface density equals the galaxy floor,

$$\eta\,\Sigma_{\rm coll}(M) = \frac{a_0}{2\pi G}, \qquad \eta = \frac{4}{3B(1)}\,c_f^{5/4},$$

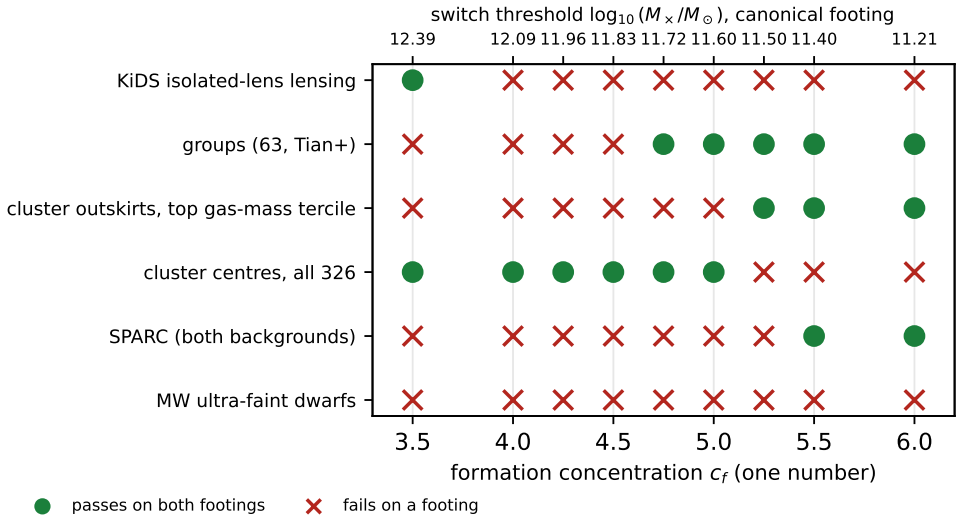
so at the external $c_f = 4$ it sits at $10^{12.09}$ (canonical) / $10^{12.43}\,M_\odot$ (alternative). X62 repaired the dwarfs but failed on groups, cluster outskirts and lensing.

3 One number cannot do both jobs (CF1)

CF1 runs X62's committed harness (SPARC on two backgrounds, 63 groups, 15 eRASS1×KiDS cluster keys in two h cells, five Local Group dwarf classes, KiDS-1000 isolated-lens lensing) with c_f as a single scalar that sets both the threshold and the cores. Step 0 reproduces X62 at $c_f = 4$ exactly; checks 10/10. Verdict: **NO SINGLE** c_f.

c_f	3.5	4.0	4.25	4.5	4.75	5.0	5.25	5.5	6.0
threshold, canonical	12.39	12.09	11.96	11.83	11.72	11.60	11.50	11.40	11.21
threshold, alternative	12.73	12.43	12.29	12.17	12.05	11.94	11.83	11.73	11.54
lensing $\Delta\chi^2$ (canonical)	0.0	+22.5	+162.7	+254.1	+325.3	+384.8	+395.7	+403.2	+416.0

Thresholds are $\log_{10}(M_\times/M_\odot)$; the lensing line passes at $\Delta\chi^2 \leq 9$.



Grid values at which each line passes on both footings: lensing 3.5 only; groups ≥ 4.75; cluster outskirts (top gas-mass tercile, G-OUT-M3) ≥ 5.25; cluster centres (G-IN-Mall) ≤ 5.0; SPARC ≥ 5.5 (the alternative footing fails it below); the Milky Way ultra-faint dwarfs never pass on the alternative footing ($Z = -2.77$ at every c_f, a dwarf-side problem).

Two tensions follow, and they are the content of this note.

1. **Threshold against concentration.** The isolated KiDS lenses must carry no halo, so the threshold must sit at or above $\approx 10^{12.4}\,M_\odot$ (canonical); groups need the concentration $c_f \geq 4.75$. In the crossover construction a higher c_f lowers the threshold, so the threshold cannot be the crossover. It must be set by something independent of halo concentration.
2. **Cluster inner against outer.** The outskirts want more concentration than the centres allow. GC1 (below) and the record's IB1 point the same way: the cluster halo's outer budget and its core concentration are separate degrees of freedom.

4 The cluster halo budget is set by measured gas (GC1, GC2, GC3, GS1)

Every tested core rule left G-OUT-M3, the outskirts of the most gas-rich clusters, under-predicted (X43 $c_f = 4$: $Z = +2.41$; X38: +5.4; IB1 with the hydrostatic gas mass: +5.02). Four frozen lanes locate the cause.

- **Not correlated structure (GC1).** Zero-parameter correlated mass beyond each cluster's turnaround radius (the record's CLASS linear correlation function, Mo–White bias, median $b = 4.1$) changes the outer predictions by at most 1.3 % (G-OUT-M3 by 0.000): it projects as a nearly flat sheet. Verdict **NOT REPAIRED**. With the native normalisation (5.2715 for 5.364) G-OUT-M3 is +2.55.
- **The budget is the lever (GC2).** The record's halo is the fair share of the baryons inside a 3 Mpc gas cut, spread to turnaround. Counting the gas to the self-consistent turnaround radius instead overshoots with the record's default gas tail (r^{-3} only beyond $2.2R_{500}$; G-OUT-M3 $Z = -4.77$) and passes with a tail steepening from R_{500}.
- **The infall shape fixes the centres (GC3).** A cored self-similar infall profile ($\rho \propto r^{-9/4}$ outside the X43 core), normalised to the fair share of the baryons measured inside R_{500}, brings the cluster centres into line with the native zoom concentration (G-IN-Mall −0.90, from −2.37), but leaves the gas-rich outskirts short (G-OUT-M3 +2.40).
- **Measured gas decides (GS1).** The X-COP gas density slope beyond R_{500} is $s = 2.777$ (bootstrap 16–84 % range 2.676–3.071; 11 clusters). Counting that measured profile to turnaround, the native fair share passes all 15 cluster keys at $c_f = 4$ on both footings (G-OUT-M3 $Z = -0.76$), with SPARC passing; only the group line fails ($Z = +2.05$). At this slope $c_f = 5$ over-predicts the clusters. The frozen verdict is therefore **MEASURED SLOPE FAILS** (no rule passes every line), with one remaining line.

The cluster outskirt problem is resolved by measured data with nothing chosen. What remains is that groups need a slightly higher concentration ($c_f \gtrsim 4.25$–4.75, CF1) than clusters take (≈ 4). X48's native zooms give c_f falling with mass (7.2 / 7.0 / 5.6 at 10^{12} / 10^{13} / $10^{14}\,M_\odot$), the direction required, though above the window and not converged.

5 Below the threshold: the exclusion mechanism (M6)

The surviving candidate for keeping cold dark energy out of galaxies below the threshold is M6: galaxies form in initial patches depleted of cold dark energy and enriched in baryons (compensated). Its decisive probe is the Lyman-α forest. Four framework-native forest models were frozen and run against Boera et al. (2019) and DESI DR1:

- P1D1 (linear proxy, z 4.2–5.0) and P1D3 (linear proxy, DESI z 2.2–4.4): not diagnostic (null χ^2 137.6/39 and 7017/719).
- P1D2 (lognormal and saturation, z 4.2–5.0): not diagnostic; saturation hides M6 there (Asimov $\Delta\chi^2$ 0.44 / 0.16).
- P1D4 (3D lognormal with redshift-space distortion, DESI): not diagnostic (null χ^2 1864/690); the $z \geq 3.6$ subset reaches 433/327.

In the regimes closest to adequate, M6 at X57's bound sits at the 95 % edge (P1D1 $k \leq 0.1$ s/km: $\mu_{95} = 1.26$; P1D3 $z \geq 3.6$: $\mu_{95} = 1.03$). M6N1 adds a derived requirement: with no exchange (X54) and X22's cap, a galaxy needing M6 must have at least 83.5 % (canonical) / 97.7 % (alternative) of its baryons patch-born. At the record's own Lagrangian retention (19.1 % / 16.3 %) M6 then needs more patch baryons than X57's provisional budget allows; it fits only if those galaxies retained at least 19–56 % (canonical) or 28–83 % (alternative) of their patch baryons.

6 The question, stated precisely

1. What physics switches cold dark energy on at $M \gtrsim 10^{12.4}\,M_\odot$ (canonical footing), independently of the concentration of the halos it then forms? Candidates must leave isolated $L_\star$ systems (including the KiDS lenses) halo-free and give groups halos.
2. Why are groups more concentrated than clusters ($c_f \gtrsim 4.25$ against ≈ 4 at the measured gas budget)? Does a native, mass-dependent concentration deliver it?
3. Below the threshold, how is cold dark energy kept out of galaxies? If by M6, what native forest model decides it, and do galaxies retain the required fraction of their patch baryons?

What would decide them: a native threshold derivation compared with the CF1 map and with a decoupled threshold map (CF2, in progress); converged, mass-dependent native zoom concentrations (the ng 64 runs are in progress); KiDS lensing and dwarf regrades at the measured cluster budget; a per-cent-level native forest model

for M6.

Provenance

Lane folders (repository `rebuild_2026_10_22/`): CF1_cf_window_map (criteria ef5a07c251, result e370b35e1c), GC1_group_cluster_core_rule (d1da6b8185, 09280e4b66), GC2_turnaround_baryon_budget (3db544593b, 8f1554e920), GC3_infall_halo_outskirts (50c3953710, 5726142b8a), GS1_measured_gas_outskirts (17064b9844, 01f6dd977d), M6N1_minimum_patch_need (3a2f5ca3a0, 3ab6a1bf44), P1D1–P1D4 (8f6e3db4bc, d50412913d, d89e0f9469, f05be7280b), X43, X58, X62. Every number above is checked against the committed lane outputs by `PAPER63_audit.py`. AI assistance: Anthropic's Claude (analysis lanes, review, drafting).